PHAM MINH QUAN

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Ho Chi Minh City, Vietnam

EDUCATION

Ho Chi Minh City University of Technology and Education (HCMUTE) **Sep 2024 – May 2026**
Bachelor of Engineering in Data Engineering *GPA: 3.0/4.0*

TECHNICAL SKILLS

Data Knowledge & Methodologies: Data Modeling (Star Schema), Medallion Architecture, ETL/ELT Pipeline Design, Exploratory Data Analysis (EDA), Data Quality, Sentiment Analysis (NLP), Machine Learning (XGBoost, LightGBM), Time Series Forecasting

Programming Languages & Frameworks: Python (Pandas, NumPy, Scikit-learn, PySpark, FastAPI), Advanced SQL, PostgreSQL, Next.js

Tools & Technologies: Power BI (DAX), dbt (Data Build Tool), Apache Spark, Apache Kafka, Apache Airflow, Delta Lake, Trino, Google Cloud Storage (GCS), Docker, Git/GitHub, MS Excel

RESEARCH & DEVELOPMENT PROJECTS

Vietnam E-Commerce Analytics Platform | *Data Analyst* *May 2026*
Personal Project (github.com/Quan-min211/ecommerce_analytics_platform)

- **Analyzed massive-scale e-commerce data** from Shopee using **Advanced SQL** to identify dynamic pricing trends, market gaps, and competitor strategies for pricing optimization.
- Designed interactive **Power BI and Next.js dashboards** to visualize core business KPIs (Customer Segmentation, Retention metrics, and Conversion proxies), enabling stakeholders to make data-driven commercial decisions.
- Extracted ad-hoc business reports and optimized data aggregations by building **Dimensional Data Models (Star Schema)** via dbt and executing complex analytical queries via Trino (SQL federation).
- Translated unstructured customer reviews into actionable business insights using **NLP Sentiment Analysis**, quantifying user satisfaction to pinpoint key drivers of product retention and customer churn.
- Guaranteed high data quality for downstream BI tools by structuring data through a Medallion Architecture (Bronze → Silver → Gold) utilizing PySpark and Delta Lake.

Real-Time Crypto Data Lakehouse | *Data Engineer* *Mar 2026*
Personal Project (github.com/Quan-min211/Crypto-DataLakehouse-Project)

- Architected a **Medallion data lakehouse** on Google Cloud Storage, processing 50 cryptocurrency trading pairs with both real-time streaming and historical batch pipelines.
- Implemented a **real-time streaming pipeline** using Binance WebSocket → Kafka → Spark Structured Streaming with 30-second micro-batches, achieving sub-minute end-to-end latency.
- Built a **3-layer ETL pipeline** in PySpark: Bronze (raw storage), Silver (deduplication, data quality rules), and Gold (OHLCV aggregations and moving average metrics).
- Orchestrated **5 Airflow DAGs** with dependency management, anti-collision locks, and self-healing retry logic, achieving fully automated data-aware scheduling.
- Containerized 10+ microservices using Docker Compose and deployed an **ML prediction dashboard** with XGBoost and LSTM models for price forecasting and anomaly detection.

LEADERSHIP & COMMUNITY SERVICE

Founder | *HCM UTE Research on Technology and Innovation Club* *May 2025 – Present*

LANGUAGES

English: Conversational Proficiency (4 skills: Listening, Speaking, Reading, Writing)